

High-Level Overview

Predictive External Data for Cement Industry (Portugal): Cement demand and financial performance are driven by construction activity and macroeconomic factors, as well as input costs and policy signals. Key external data fall into several categories: **(1) Construction demand indicators** (e.g. building permits, construction output, cement consumption) which signal future sales volumes ¹ ²; **(2) Real estate and credit metrics** (housing permits, property prices, mortgage lending) reflecting investment cycles and funding conditions ³; **(3) Industry supply & competition data** (cement production/consumption, import/export volumes) indicating market share and competitive pressure ²; **(4) Cost driver indices** (energy prices, fuel costs, CO₂ allowance prices, transport and raw material cost indices) that heavily impact margins ⁴ ⁵; **(5) Public investment and policy signals** (infrastructure tender awards, government/EU investment plans) which create future demand for cement and concrete ⁶; **(6) Other high-value signals** like **weather patterns** (rainfall, extreme heat) affecting short-term construction activity ⁷, and **economic sentiment surveys** (construction confidence indexes) that provide early indication of sector trends. Integrating these diverse external sources into forecasting models can greatly enhance the prediction of a cement company's volumes, revenues, and costs by providing *leading indicators* (e.g. permits, sentiment), *coincident measures* (e.g. production indexes), and *cost foresight* (e.g. energy, carbon prices).

Driver-to-Data-Source Map

Driver Category	Financial KPIs Affected	External Data (Examples)	Priority
Macro & Construction Demand	Cement & concrete sales volumes; Revenue	Building permits, construction output index, cement consumption	High – Core demand drivers ¹ ²
Real Estate & Credit Cycle	Cement volumes (housing segment); Pricing power	Housing permits, house price indices, mortgage volumes & rates	High – Housing drives cement demand; credit leads investment ³
Sector Supply & Competition	Market share, pricing pressure, export sales	Cement production & consumption, import/export volumes, capacity data	Medium – Context for competition; moderate direct predictiveness
Cost Drivers (Energy, CO₂, etc.)	Fuel costs, power costs, CO ₂ costs; Margins	Electricity price, coal/petcoke price, diesel price, CO ₂ EUA price, construction material cost index, wage index	High – Major impact on cost lines and margins ⁴ ⁵
Public Investment & Policy	Future cement demand (infra projects); Order backlog	Public works tenders & contracts, government/EU infrastructure budgets, EU fund disbursements	Medium – Significant demand signals (infra) but data integration is complex ⁶

Driver Category	Financial KPIs Affected	External Data (Examples)	Priority
Weather & Seasonality	Short-term sales fluctuations; Project delays	Rainfall, temperature, extreme weather days (by region/season)	Medium – Affects monthly output (esp. winter rains) ⁷ ; vital for short-term forecasts
Economic Sentiment & Other	Overall demand climate; Downside risks	Construction confidence index, PMI, consumer sentiment, etc.	Low/Med – Early signals of trend shifts; useful for context ⁸

(Priority reflects a combination of predictive value for financial forecasting and importance for initial integration.)

Detailed Data Source Catalogue for Portugal

Below is a catalogue of specific external data sources/datasets in Portugal, each with a profile for practical use. (All sources are **free/open** unless noted; paid sources are mentioned as optional.)

Macro & Construction Demand Indicators

- **Source ID:** PT_INE_PRODCONST
- **Provider:** Statistics Portugal (INE)
- **Dataset/Portal Name:** *Índice de Produção na Construção e Obras Públicas* (Production Index in Construction and Public Works)
- **Link:** [INE database – Construction Production Index \(monthly\)](#) (via INE API or website)
- **Description:** Index measuring the *volume of construction work done*, split into Building Construction and Civil Engineering sub-indices. It is a seasonally and calendar-adjusted volume index (base year 2015=100 or 2021=100 in latest series). A higher index means more construction activity performed.
- **Key Variables:** Monthly index values and year-over-year growth for **Total Construction Output**, and by sub-sector (**Buildings** vs **Civil Engineering**). e.g. April 2024 index = 112.8 (seasonally adjusted), up 3.4% YoY ⁹ ¹⁰ .
- **Geography:** National total (Portugal); possibly available broken down by broad sector but not by region (the index is country-wide).
- **Time Granularity & History:** **Monthly** data. Long history available (the series is continuous for decades with base updates). Data from at least the 1990s (older base series can be linked/extended).
- **Update Frequency & Lag:** **Monthly**, published ~6 weeks after month-end. (Example: April data released mid-June ¹¹ .)
- **Access Method:** INE API (JSON) – available with indicator code; also downloadable via INE's online database (CSV/Excel). The INE API is open and allows automated retrieval (with a 40k data-point per call limit) ¹² .
- **Licensing/Cost:** Free, open official data. No restrictions for use (attribution to INE is courteous).
- **Predictive Use Cases:** A **coincident** indicator for construction activity – useful to **nowcast** cement and concrete sales in the same month ⁹ . Can serve as a benchmark for total construction output against which company sales can be compared. Can also be lagged to see persistence (e.g. a sustained rise in this index indicates robust demand environment).

- **Leading/Lagging:** *Coincident* for construction volumes (with minimal lead/lag). It's a contemporaneous measure of sector output; however, trends in this index may lead financial results slightly since it is published before company financials.
- **Predictive Relevance:** 4/5 – Directly reflects the overall construction market health (which drives cement demand). However, it is somewhat broad (includes building and civil; not specific to cement consumption) and is more useful for tracking current momentum than forecasting big turns (those are better caught by permits or confidence).
- **Ease of Integration:** 5/5 – **High**. The INE API and data portal make it straightforward to pull this series regularly. The series is well-defined and requires simple time series handling.
- **Comments/Caveats:** Ensure to use the calendar-adjusted series for an accurate measure (INE provides both raw and adjusted data). Revisions can occur (minor) when base years are updated or seasonal adjustment is refined. The index is national; regional demand patterns would not be visible here.
- **Source ID:** PT_INE_BLD_PERMITS
- **Provider:** Statistics Portugal (INE)
- **Dataset/Portal Name:** *Obras Licenciadas e Concluídas* (Building Permits and Completed Works) – **Building Permits** sub-dataset
- **Link:** [INE – Construction: Permits and Completions](#) (also via INE API)
- **Description:** Official data on **construction permits** issued by municipalities. It covers the number of **new building projects licensed** and also data on completed buildings. Key metrics include number of licensed buildings, number of dwellings in those permits, and licensed construction area. Permits are a forward-looking indicator as they precede actual construction work.
- **Key Variables:**
 - **Buildings licensed** (count), broken down by type (residential, non-residential).
 - **Dwellings licensed** (count of new housing units approved).
 - **Floor area licensed** (m²). Data often reported as quarterly totals, but monthly data are collected (as evidenced by monthly YTD figures). For example, **30,615 new dwellings** were licensed in Jan–Sep 2025, up 21.1% YoY ¹³. The number of building permits (all types) was up ~9.9% YoY up to July 2025 ¹. This shows a strong leading growth signal in permits.
- **Geography:** Available at **national** level and by region (NUTS II, possibly municipality in detailed data). INE releases national summaries; the data portal allows regional breakdowns.
- **Time Granularity & History:** Collected **monthly**, but often published as **quarterly** aggregates in official reports. Long history (permits data extends back many years; e.g., data since 1990s in some form).
- **Update Frequency & Lag:** **Quarterly releases**, typically ~6–8 weeks after quarter's end (e.g., Q3 data released mid-Nov). Internally, monthly figures can be accessed via the database/API soon after quarter close. **Monthly** series can be constructed from the API. Lag is moderate – permit counts for, say, September become available by early November.
- **Access Method:** INE API/database. The INE database allows filtering by "Reference period" (month/quarter) and geographical level. Data can be downloaded as CSV/Excel. No formal API for live updates beyond this database API (which is JSON-based).
- **Licensing/Cost:** Free official data. No use restrictions (public domain government data).
- **Predictive Use Cases:** A **key leading indicator** for future **cement and ready-mix demand**, especially in the **building construction** segment. Building permits signal upcoming construction starts; historically, a rise in permits leads to higher cement consumption in ~3–9 months as projects

break ground. For example, a surge in housing permits usually leads to increased concrete and cement sales as those projects progress. We would use permit data to forecast **sales volumes** by region/product (more housing permits → more bagged cement and ready-mix demand). Also useful for **capacity planning** (permits downturn might warn of a sales dip ahead).

- **Leading/Lagging Behavior:** *Leading.* Permits often lead actual construction activity by several months. E.g., developers obtain permits a few months (for residential) up to a year (for complex projects) before material consumption. Empirically, **building permits are a strong leading indicator (~6 months lead)** for cement demand ¹. We might use moving averages or lagged values of permits in regression models to predict future sales.
- **Predictive Relevance:** 5/5 – **Very High.** Permits data is highly relevant in Portugal's context because housing construction is a major consumer of cement. It captures future **demand pipeline**. The growth in licensed dwellings (e.g. +21% YoY in 2025 ¹³) directly foreshadows strong market demand.
- **Ease of Integration:** 4/5 – **High.** Data is accessible via INE's API, but because official publication is quarterly, we might need to extract monthly figures from the database. Automation is feasible. Minor complexity: ensuring the correct aggregation (monthly vs quarterly) and aligning it with our forecasting periods.
- **Comments/Caveats:** Pay attention to differences between **permits for new construction vs rehabilitation** – new construction permits have the strongest relationship with cement demand. Also, regional distribution of permits can help regional sales forecasts (if available, we may integrate by matching regions to our sales territories). The data can be volatile month-to-month; using a 3-month moving average or YoY growth rates can smooth seasonality. Additionally, a permit does not guarantee construction starts (some permitted projects may be delayed or cancelled), but in aggregate it's a reliable demand proxy.

- **Source ID: PT_INE_HOUSING_IDX**

- **Provider:** Statistics Portugal (INE)
- **Dataset/Portal Name:** *Índice de Preços da Habitação (IPH)* – Housing Price Index
- **Link:** [INE – House Price Index](#) (also in Eurostat database)
- **Description:** Quarterly index of **residential property transaction prices** in Portugal. Reflects market prices for housing (both new and existing homes). It is a lagging economic indicator but a vital measure of the real estate cycle, which underlies construction demand. Rapid house price increases often stimulate new development, whereas price drops can signal a cooling market.
- **Key Variables:** House Price Index (2015=100 or 2010=100 depending on base) – overall, and breakdown by **new dwellings vs existing dwellings**. Also available: year-on-year % change. For example, recent figures (Q2 2025) show continued growth in house prices ~7-10% YoY (hypothetical example). This indicates strong real estate market momentum.
- **Geography:** National index. (Also available by region – e.g., NUTS2 indices – through data portal or Eurostat.)
- **Time Granularity & History:** **Quarterly** (Q1 2009 onward for IPH series). Historical data for earlier years may be available via older series or other sources (e.g., estimates back to 1996 in some publications).
- **Update Frequency & Lag:** **Quarterly**, published ~70 days after quarter-end (e.g., Q2 released mid-September). Significant lag relative to real-time market conditions.

- **Access Method:** INE website (press releases, Excel) and **Eurostat** API (the IPH for PT is included in Eurostat's housing price database). Also, INE's API can provide the index by selecting the indicator code.
- **Licensing/Cost:** Free official data. Eurostat/INE data is open for reuse.
- **Predictive Use Cases: Contextual/leading indicator for construction demand** – surging house prices can encourage more construction (developers respond to price signals). The index helps predict **volume of future housing projects** (if prices keep rising, more projects become viable, boosting permit numbers and cement demand with a lag). Also used in **affordability analyses** that might feed into demand forecasting (if prices + interest rates outrun incomes, housing demand could slow). For financial forecasting, house prices can inform revenue forecasts indirectly: a strong housing market might correlate with better pricing power for cement or higher volumes.
- **Leading/Lagging Behavior:** *Lagging* relative to construction activity (prices often continue rising even after construction peaks, and vice versa). However, it's *leading* relative to **housing permits** (developers need sustained price rises to initiate projects). So one could say it is a moderately **leading indicator for permits and construction starts** on a longer horizon (6–12 months).
- **Predictive Relevance:** 3/5 – **Moderate**. While not directly tied to cement sales in the short term, it captures the macro environment of housing. It's more useful for **medium-term scenario planning** (e.g., if house prices flatten or fall, anticipate a slowdown in new construction 1–2 years out). Less directly predictive than permits or credit, but still valuable for broader models.
- **Ease of Integration:** 5/5 – **High**. Available via standardized sources (Eurostat API) in a stable format. Quarterly updates are infrequent, easy to maintain.
- **Comments/Caveats:** House prices can be influenced by many factors (interest rates, supply constraints) and may not translate one-to-one into construction volumes (e.g., if land or licensing is a bigger bottleneck). Also, consider using **median house price or hedonic indices** if available for more granular insight; **Confidencial Imobiliário** (see Tier 2) provides detailed city-level indices (but requires subscription). For our national-level forecasting, the official HPI suffices as a barometer.
- **Source ID:** PT_INE_BANKVAL
- **Provider:** Statistics Portugal (INE)
- **Dataset/Portal Name:** *Valor Mediano de Avaliação Bancária* – Median Bank Appraisal Value (housing)
- **Link:** [INE – Housing Bank Appraisal Value](#)
- **Description:** A **monthly indicator** of housing prices based on **banks' appraisal values per square meter** for mortgage purposes. This complements transaction price indices by providing **up-to-date insight** into property valuation trends. Banks evaluate properties for new mortgages, so this captures price trends for both new and existing houses on a monthly basis.
- **Key Variables:** Median appraisal value (€/m²) of housing at the national level (and by region, and by apartment vs house). For example, in Sep 2025 the median appraisal was **€1,995/m²**, up 17.7% YoY ¹⁴. The index often shows continuous monthly increases when the market is hot.
- **Geography:** National and **regional (NUTS III)** breakdown available (e.g., Lisbon, Norte, Algarve etc.). Regional data is useful for identifying hotspots where construction might concentrate.
- **Time Granularity & History:** **Monthly**, with data available from around 2011 onward (since this series was introduced around that time).
- **Update Frequency & Lag:** **Monthly**, published ~one month after month's end (e.g., September data released in late October). Very short lag – a timely indicator.
- **Access Method:** INE API (indicator code for median appraisal by region), or monthly press releases (Excel). Data can be scraped or pulled via JSON.

- **Licensing/Cost:** Free. No restrictions.
- **Predictive Use Cases: Leading indicator for housing construction demand.** Rapid rises in appraisal values indicate strong housing demand and likely spur more construction (developers see upside). We can use it to refine short-term forecasts of cement demand in the housing segment – e.g., if appraisals plateau or drop for a couple of months, it may predict a slowdown in new project launches. It's also useful for **credit risk context** (if values drop, banks may lend less, impacting construction financing).
- **Leading/Lagging:** *Leading*, relative to construction activity, by perhaps 3–6 months. It captures market sentiment almost in real-time. It might even serve as a **coincident indicator for sales** (if housing market cools, cement orders from residential projects might see a dip shortly after).
- **Predictive Relevance:** 4/5 – High. Particularly relevant in a housing-driven market – it's a proxy for **real estate health** and thus future cement consumption. However, it's an indirect indicator (of prices, not volumes), so we treat it as supplementary to permits.
- **Ease of Integration:** 5/5 – Very easy. Monthly updates and available through the same INE API. Small data size.
- **Comments/Caveats:** Being a median of appraisals, it could be influenced by the mix of properties financed (e.g., more high-end homes appraised that month). But large sample sizes typically even that out. Use year-over-year % changes or 3-month averages to spot trend inflections. We should also watch out for policy changes (if any) in appraisal methods or lending standards as they could shift the series.
- **Source ID: PT_INE_CONF_CONSTR**
- **Provider:** Statistics Portugal (INE) / European Commission (DG ECFIN)
- **Dataset/Portal Name:** *Índice de Confiança da Construção e Obras Públicas* – Construction Confidence Indicator (from business surveys)
- **Link:** [INE – Monthly Economic Survey Press Release](#) (contains confidence indices; also available via [European Commission's dataset](#))
- **Description:** A monthly sentiment index from a survey of construction firms, summarizing managers' expectations for activity, order books, and employment. Expressed as a balance of positive vs negative responses (seasonally adjusted). This is a well-known **leading indicator** for the construction cycle, capturing industry sentiment before hard data reflects changes.
- **Key Variables:** Construction Confidence Index (unitless, typically an average=0 balance). For example, a value of **-10** might indicate overall pessimism, while +5 would indicate optimism. In mid-2025, this indicator showed improvement, then slight dips – e.g., it **increased in May-June 2025 after falling in prior months, reflecting better expectations** ¹⁵. The exact values are often negative in construction (structural pessimism bias), but *trends* (rising or falling) are telling.
- **Geography:** National (Portugal). No regional breakdown (the survey is aggregated nationally).
- **Time Granularity & History:** **Monthly**, long history (the European Commission has series since the 1980s for Portugal's construction confidence).
- **Update Frequency & Lag:** **Monthly**, usually released at end of the same month (e.g., October survey published in late October). Very prompt.
- **Access Method:** The European Commission's BCS (Business and Consumer Survey) database offers CSV downloads and an API. INE publishes the headline in a press release (with other climate indicators) and possibly in the INE API as an indicator. For automation, using the EC's data (which includes Portugal's survey) is convenient.
- **Licensing/Cost:** Free (EU data, open license).

- **Predictive Use Cases: Early-warning indicator** of trend changes in construction activity. Useful for **short to medium-term forecasting**: if confidence drops sharply for several months, one can expect slower orders and possibly lower cement sales in upcoming months. Conversely, rising confidence often precedes upticks in permits and investment. For forecasting models, this index could be an exogenous predictor variable (leading by ~1-3 months) for cement sales or revenue. It can also help qualitatively in adjusting forecasts (e.g., if all hard data is good but confidence plunges due to some shock, we might temper near-term outlook).
- **Leading/Lagging: Leading.** It is designed to lead actual output. The survey asks about order books and expected activity, so it moves **before** production and sales volumes. Typically provides a 1-quarter lead on construction output fluctuations.
- **Predictive Relevance: 3/5 – Moderate.** While it has predictive value, especially for turning points, it is an indirect measure. We would not rely on it alone, but in combination with permits and other data it adds robustness (especially when there's a divergence – e.g., permits high but confidence falling might signal execution or financing issues).
- **Ease of Integration: 5/5 – Very easy.** Data is readily accessible via EU databases. Updates are timely and consistent.
- **Comments/Caveats:** The confidence indicator is qualitative and can be volatile month-to-month. It's best analyzed with smoothing or looking at trends. It may also react to factors like political uncertainty that don't always materialize into changed construction volumes. Use it as a barometer rather than a precise measure. Also note it covers **all construction** (building + civil), so a big infrastructure sentiment change could move it even if housing is stable (and vice versa).

• **Source ID: PT_ATIC_CEMENT**

- **Provider:** ATIC – *Associação Técnica da Indústria de Cimento* (Portuguese Cement Industry Association)
- **Dataset/Portal Name:** *Consumo de Cimento no Mercado Nacional* (Cement Consumption – Portugal)
- **Link:** *No public direct link* (data is published via industry reports such as AICCOPN's bulletins ¹⁶ and press releases). ATIC is the source behind these figures.
- **Description: Monthly total cement consumption in Portugal**, measured in metric tons. This effectively equals domestic cement sales (production + imports – exports). It is a critical industry metric, effectively the *target variable* we want to forecast (aggregate demand). ATIC gathers data from member companies (major cement producers Cimpor and Secil) and possibly customs, to report the total market consumption.
- **Key Variables: Cement consumption volume** (usually reported as thousands of tonnes) for each month and cumulative year-to-date. E.g., *September 2025 YTD cement consumption = 3.039 million tonnes, up 0.2% YoY* ¹⁷. Monthly values: e.g., **February 2024: 646 thousand tonnes, +13.4% YoY** ². Growth rates (monthly or YTD) are also reported. This can be broken by domestic vs export if needed, but usually it's total domestic market.
- **Geography:** National total (Portugal). No regional breakdown provided publicly.
- **Time Granularity & History: Monthly** totals. History – ATIC likely has long historical series (decades). Publicly, we see references to year totals and YoY changes in press going back many years (some sources like Cembureau compile annual country data too).
- **Update Frequency & Lag: Monthly**, usually available ~15-30 days after month-end. (For example, AICCOPN's November 2025 bulletin gave data up to September 2025, which was available by mid-November ¹⁸. Often data up to month M is out by M+1 or M+2 at latest via industry channels.)
- **Access Method:** There's no open API. Data is published in reports: **AICCOPN's "Síntese Estatística da Habitação"** includes the cement consumption YTD figure monthly ¹⁶. Press releases via **trade**

magazines (e.g., CemNet, Construção Magazine) also report these figures ². To integrate: one approach is to scrape those monthly bulletins or collaborate with ATIC for a direct feed (since we're an industry stakeholder, we might arrange to get the data via email or members' portal). Alternatively, since ATIC has only two main members, the company itself might have internal aggregate numbers (if we're a producer). But for completeness, we treat ATIC's data as the official external source.

- **Licensing/Cost:** Essentially free for industry use, but not openly published in a machine-readable format. ATIC is an industry association; data is shared publicly in summary form at no cost. If more detail is needed, membership or contacts might be required. No known legal restrictions on using the summary numbers.
- **Predictive Use Cases:** This is *the measure of market demand* – we would use historical cement consumption to benchmark our company's sales vs market and to model seasonality and growth trends. It's also useful as a target for forecasting overall market size. In multivariate models, we might use other external drivers to predict this series itself. For internal forecasts, we might prefer to predict our own sales, but having the total market helps gauge **market share changes** and sanity-check forecasts. Because it's aggregate demand, many leading indicators (permits, PMI, etc.) will be used to predict **this**. In scenario analysis, one could forecast cement consumption under different assumptions and then derive our share.
- **Leading/Lagging:** *Coincident* – it is effectively the outcome variable (realized demand). However, from a company perspective, having the total market data might lag our own sales data (which we get in real-time). So in operational terms, ATIC data often arrives a bit later (a few weeks into the next month) and is useful for validation rather than advance prediction. Nonetheless, trends in ATIC's data combined with our sales can give insight into how demand is evolving beyond our customer set.
- **Predictive Relevance:** 5/5 (for market tracking) but 1/5 as a *driver* since it *is* the outcome. In context, we treat it as **Tier 1 reference data** rather than a predictor. It's crucial for forecasting because it's the ground truth of demand that our other indicators aim to predict.
- **Ease of Integration:** 2/5 – **Medium/Low**. There's no official automated source. We likely need to retrieve it from PDF reports or press articles each month. This might involve building a small scraper for AICCOPN's site or subscribing to industry newsletters. Once obtained, it's straightforward numeric data. Automating it is the main challenge.
- **Comments/Caveats:** Minor discrepancies can occur if a producer revises figures. Also, caution if interpreting short-term fluctuations: e.g., a big year-on-year jump might be due to an unusually low base last year (weather or strikes). But overall, this is the definitive metric for demand in Portugal. We should also consider *clinker and cement export* data separately if exports are significant (though for financial forecasting of domestic business, domestic consumption is key). Also note ATIC's involvement means data is likely trustworthy as it comes from the companies themselves.

Real Estate & Credit Indicators

- **Source ID:** PT_BdP_CREDIT_HOUSING
- **Provider:** Banco de Portugal (Central Bank)
- **Dataset/Portal Name:** *BPstat – Monetary and Financial Statistics*, specifically **New Loans for House Purchase** and **Housing Credit Stock**
- **Link:** [BPstat Statistics Portal](https://bpstat.bportugal.pt/data/docs) (Data domain: "Monetary and Financial Statistics > Loans > New loans by purpose") – API available at `bpstat.bportugal.pt/data/docs`
- **Description:** Data on mortgage lending which signals housing market financing conditions. This includes the monthly amount of **new housing loans granted** by banks, and the total **outstanding**

stock of mortgage credit. High growth in new housing credit indicates strong demand for housing (and ability to finance projects), which can drive construction. Conversely, a credit slowdown can foreshadow a dip in construction activity.

- **Key Variables:**

- **New housing loans (flow):** e.g., €2.321 billion in Jan–Feb 2024, +26.1% YoY ³. For full-year 2024 this might be much higher (the AICCOPN report showed **€16.884 billion in new housing credit in Jan–Sep 2025, +37.1% YoY** ¹⁸ ¹⁹). Typically measured monthly in € millions.
- **Housing credit stock:** total outstanding mortgage debt (e.g., ~€108 billion as of Sep 2025 ²⁰), with YoY growth (e.g., +7.2%).
- Possibly **credit to construction/real-estate companies:** credit stock to construction and real estate sectors (sometimes reported separately, e.g., ~€16.5B stock to Construction & Real Estate companies, +4-5% YoY ²¹). This indicates banks' funding to developers and builders.
- **Interest rate on new housing loans:** average interest % (e.g., 3.23% in Sep 2025, down 113 bps YoY as per AICCOPN ²² ²³). Rising rates can dampen future loan demand.
- **Geography:** Nationwide (Portugal). No regional breakdown in central bank data.
- **Time Granularity & History: Monthly** (for flows and interest; credit stock often monthly too). History likely back to 2003 or earlier in some form; certainly since Euro adoption (1999) for interest rates.
- **Update Frequency & Lag: Monthly**, published ~1 month after period (e.g., October data in late November, as part of BoP/Monetary stats release). The central bank is usually timely.
- **Access Method:** Banco de Portugal provides an API and bulk downloads via BPstat. We can query specific series by codes (the GitHub scripts and documentation ²⁴ show how to use it). Alternatively, Eurostat and ECB Statistical Data Warehouse also carry some Portuguese banking stats. The data is machine-readable in multiple formats (JSON, Excel from portal).
- **Licensing/Cost:** Free (official stats). Data can be used with attribution.
- **Predictive Use Cases: Leading indicator for housing-related cement demand.** If *new mortgage issuance* is surging, it reflects households buying homes (often new or off-plan) – which correlates with more construction and sales of cement/ready-mix to residential projects. It also reflects developer financing conditions (easier credit = more projects launched). We'll incorporate mortgage volumes as a predictor for **future sales volumes in the residential segment**. Also, **interest rates** have an inverse leading effect: rising rates may cool housing demand with ~6-12 month lag, thus forewarn slower cement demand growth. These data also feed into revenue forecasts indirectly (if housing market slows, perhaps pricing gets competitive).
- **Leading/Lagging: Leading.** New loan volumes often lead construction activity by a few months – as people secure financing before purchase/build, and developers secure loans before starting projects. Mortgage approvals can also coincide with sales of newly completed homes, which is a bit lagging relative to construction start. But overall, credit growth is a **leading economic indicator**. Interest rate trends are strongly leading (rate hikes can curtail future demand).
- **Predictive Relevance: 4/5 – High.** In Portugal, the housing market is credit-driven; big swings in mortgage volumes have historically preceded booms/busts in construction. For example, the data shows mortgage issuance boomed ~37% YoY in 2025 ¹⁹, aligning with a construction upswing – extremely relevant to forecast sustainability of demand. One could argue it's indispensable for anticipating turning points.
- **Ease of Integration: 5/5 – High.** The central bank's data portal is robust and the API is well-documented. We can automate monthly pulls for the specific series. Data are consistent and subject to minor revision only.

- **Comments/Caveats:** We should adjust for inflation (credit in nominal euros may rise partly due to house price inflation – but since construction volumes relate to real activity, we may consider real terms). Also watch policy changes: e.g., any macroprudential restrictions on lending (loan-to-value caps, etc.) can sharply impact loan volumes. Such context should accompany interpretation of the numbers.

• **Source ID:** PT_BdP_CREDIT_CONSTR

- **Provider:** Banco de Portugal
- **Dataset/Portal Name:** *BPstat – Credit to Corporates by Sector*, specifically **Loans to Construction and Real Estate Companies**
- **Link:** [BPstat Sectoral Credit data](#) (under “Loans to Non-Financial Corporations by branch of activity”)
- **Description:** This dataset provides the outstanding credit to companies in the construction sector (and real estate sector). It serves as an indicator of financial support to the industry. If banks increase lending to construction firms, it may enable more projects (leading indicator). Conversely, deleveraging or rising bad loans in this sector could signal stress and reduced capacity to invest.
- **Key Variables:**
 - **Credit stock to Construction & Real Estate** (we might combine or separate these NACE categories): e.g., ~€16.5 billion as of Sep 2025 for Construction & Infrastructure (as per AICCOPN, labeled “C&I”) ²¹, slightly up YoY.
 - **Non-performing loans ratio** in construction (if available) could also be insightful, indicating sector health.
 - **New loans to construction companies** (if the central bank breaks it down) – possibly available but less frequently cited.
- **Geography:** National, by sector.
- **Time Granularity & History:** **Monthly** or **Quarterly** (some sector credit stats are quarterly averages). History: since early 2000s.
- **Update Frequency & Lag:** Usually **monthly**, published with similar lag as other credit data (~T+1 month). Some breakdowns might be quarterly.
- **Access Method:** Same BPstat API/portal. Possibly need to filter by NACE code for Construction (F) and Real Estate (L). Data likely in millions of euros.
- **Licensing/Cost:** Free.
- **Predictive Use Cases:** Used to gauge **investment capacity in the industry**. For instance, a sustained increase in credit to construction firms might foreshadow more aggressive project development (leading to higher cement demand). If credit to the sector is contracting, it could constrain growth irrespective of demand (companies can’t fund new projects). It also can be a risk indicator – rapid buildup of debt might precede a downturn or indicate overheating. As a forecasting input, it’s somewhat macro-financial: we’d use it to adjust medium-term scenarios (e.g., if credit to construction has been shrinking for a year, perhaps the boom is running on internal funds and could slow).
- **Leading/Lagging:** *Leading*, but with a caveat – it can be both a cause and effect. Often credit expands as activity expands. It might not lead permits (could be coincident). However, credit tightness clearly *lags* a downturn (banks pull back after trouble starts). So interpret with context: it’s a good concurrent indicator of the financial cycle affecting construction.
- **Predictive Relevance:** 2/5 – **Low-Medium**. It’s useful background, but less directly predictive than mortgage data or permits. In a stable environment, it won’t change dramatically month to month. Its

value shines when there are credit crunches or booms. So we monitor it especially for warning signs of a credit squeeze that could derail forecasts.

- **Ease of Integration:** 5/5 – Easy via BPstat.
- **Comments/Caveats:** Construction and Real Estate are sometimes combined in banking stats; ideally we separate pure construction (which finances contractors) from real estate (developers). Also note that some construction activity might be financed via corporate loans not classified neatly or via bond markets, etc., but bank lending is dominant for SMEs. We should be aware of any write-off or reclassification that could cause jumps (for example, a large developer's loans moving categories).
- **Source ID:** PT_CI_RE_PRICE (Paid/Optional)
- **Provider:** Confidencial Imobiliário (CI) – private real estate data firm
- **Dataset/Portal Name:** *Índice de Preços Residenciais CI* (Residential Real Estate Index by region/city)
- **Link:** [Confidencial Imobiliário Índices](#) (subscription required for detailed data)
- **Description:** High-granularity real estate price indices and market indicators (e.g., residential prices by municipality, rents, yields, transaction counts). CI collects data from listings and transactions. This is **optional** for deeper analytics: e.g., city-level house price trends or permit counts complementing official stats.
- **Key Variables:** e.g., **Lisbon house price index** (monthly, showing, say, +7% YoY), **Rental index**, **Sales volume (# of transactions)** in major cities, etc. These can help a cement company understand local market dynamics (where growth is hottest).
- **Geography:** City/Metropolitan level (Lisbon, Porto, Algarve, etc.). Also aggregates like national excluding Lisbon.
- **Time & Frequency:** **Monthly** indices. Data available since 2007 for some series. Typically published a few weeks after month-end to subscribers.
- **Access:** **Paid.** CI requires subscription or purchase. Some high-level numbers are shared in press releases or via their platform. No open API without subscription.
- **Use & Predictive Value:** It's a **nice-to-have**: If our business is concentrated in certain regions, these indices might improve regional demand forecasts. For example, if Porto's prices are surging ahead of the national average, cement sales in North region might outperform. It can refine our permit-based forecasts by adding market sentiment. However, for high-level financial forecasting, the official national stats suffice; CI data would be more useful if we later build a detailed regional forecasting model or need to validate official data.
- **Priority:** Lower (Tier 2) due to cost and because INE's free data covers many needs.
- **Integration:** If obtained, data comes via Excel/CSV from CI; integration would be manual or via their API (if available to clients).
- **Caveats:** Ensure licensing allows internal use. And CI data might not align perfectly with INE definitions (but generally trends correlate).

Sector Supply, Trade & Competition Data

- **Source ID:** PT_INE_TRADE_CEMENT
- **Provider:** Statistics Portugal (INE) / Eurostat Comext
- **Dataset/Portal Name:** *International Trade in Goods – Cement (HS 2523)*
- **Link:** [INE International Trade Database](#) (or [Eurostat COMEXT](#) for commodity-level trade)

- **Description:** Data on **cement imports and exports** (by volume and value). Cement companies should track this to gauge **competition from imports** and foreign market demand. For instance, if imports of cement into Portugal rise significantly, it could pressure local producers' sales (and prices). Similarly, export volumes reflect external demand or surplus production.
- **Key Variables:** Monthly or quarterly **import volume** (in tonnes) of hydraulic cement into PT, and **export volume** of cement out of PT. Also their values (€) to compute unit values (approx price). For example, imports might be X thousand tonnes/month, usually low but potentially growing if domestic price > import cost. Exports might be sizable if local producers serve Spain or other markets. (E.g., 2024: Portugal imported ~**0.2Mt** and exported ~**0.5Mt** of cement – hypothetical figures).
- **Geography:** National total (trade crossing border). Can be broken down by trading partner country (Eurostat provides by origin/destination).
- **Time Granularity & History:** **Monthly** trade data. History: since at least early 2000s in detailed trade stats.
- **Update Frequency & Lag:** **Monthly**, published ~6 weeks after month's end (trade stats lag). So November imports/exports come mid-January.
- **Access Method:** Eurostat's COMEXT database (API or bulk CSV) can be used by specifying product code (HS 2523 or CN8 codes for cement types) and Portugal. INE also disseminates some high-level trade by product in their database. Integration via Eurostat API is straightforward.
- **Licensing/Cost:** Free (official trade data).
- **Predictive Use Cases: Market share and supply balance** analysis. We use this data to adjust our volume forecasts: if imports are trending up, our sales might face headwinds (especially on price). For example, cheap cement from Spain could cap our ability to grow volumes even if local demand is up. It's also useful for **pricing strategy** – if import volumes surge when local prices rise, it suggests a ceiling to pricing power. On the cost side, export data can indicate if local producers are running kilns at high utilization (strong exports often mean strong production). For forecasting financials, this is more about **sensitivity and scenario**: e.g., assume imports remain at 5% of consumption vs scenario where they double to 10%.
- **Leading/Lagging:** *Coincident/lagging*. Trade reacts to domestic supply-demand gaps and price differentials. It might not lead domestic demand, but a sudden import increase could immediately impact our sales (coincident). It's important for explaining *actuals* and adjusting forecasts rather than predicting ahead by long periods. However, if we monitor cement prices regionally, we might predict that sustained price differences will lead to future import changes – but that's a secondary inference.
- **Predictive Relevance:** 3/5 – **Moderate**. Not a primary driver of demand, but crucial for forecasting *market share and price*. If our financial performance depends on capturing a certain share, imports are effectively a subtraction from our accessible market. In a stable scenario, imports may be negligible; but any trend change here is significant. Thus, it's a **monitoring indicator** in our model.
- **Ease of Integration:** 4/5 – High. Eurostat API can be set up to retrieve monthly trade volumes by HS code. Data volumes are small. Just need mapping from HS code to "cement" (which is straightforward: HS 2523 covers cement).
- **Comments/Caveats:** The quality of trade data is generally good, but minor issues: re-exports can blur numbers (unlikely big for cement). Also, watch for **clinker vs cement** if separate codes; if clinker imports rise, it could indicate grinding activity increases. Our focus is finished cement. Additionally, consider *competitor behavior*: if a new competitor plant opens just across border, import data will catch resulting flows – we should then investigate cause. But such intelligence might come from industry news even earlier.

- **Source ID:** EU_CEMBUREAU_STATS (Optional)
- **Provider:** Cembureau (The European Cement Association)
- **Dataset/Portal Name:** *Annual Production & Consumption – by country (EU)*
- **Link:** [Cembureau Activity Report](#) (annual PDF)
- **Description:** Cembureau publishes aggregated data for cement production, consumption, and capacity for European countries annually ²⁵. This can be used for benchmarking Portugal against others, and checking official figures in one place.
- **Key Variables:** Annual cement production (Mt), cement consumption (Mt) – e.g., Portugal consumed **3.90 Mt in 2023, +1.8% YoY** ²⁶. Also capacity and consumption per capita.
- **Geography:** Country-level (Portugal and other EU).
- **Time & Frequency:** **Annual**, released mid-year for prior year. Historical trend often given for last 5–10 years.
- **Access:** Free report, but not machine-readable; manual extraction from PDFs.
- **Use:** Good for strategic context and long-term scenario building (e.g., trend growth, peak vs trough comparisons). Not used in short-term forecasting due to low frequency.
- **Priority:** Low (Tier 2). Use as reference, not for monthly model input.
- **Caveats:** Ensure consistency with ATIC data (they should align; any major discrepancy would need investigation).
- **Source ID:** PT_DGEG_PROD_CAP (Potential internal data or regulator)
- **Provider:** DGEG – Direção-Geral de Energia e Geologia (Ministry) and ATIC (for capacity)
- **Dataset Name:** *Cement Production & Clinker Capacity* (industrial stats)
- **Link:** [DGEG Mineral Statistics](#) (if available; not directly confirmed)
- **Description:** If available, data on **domestic cement production volumes** (monthly) and plant capacity utilization. In some countries, the national authority or industry reports production. In PT, ATIC likely has production data (which combined with consumption and trade can be derived). Knowing production helps for forecasting **cost structure** (if production falls, fixed costs per ton rise) and for inferring inventory changes.
- **Use:** Secondary – we can derive production = consumption – net imports. If needed, could integrate monthly production index for “cement, lime and plaster” from industrial production stats (INE publishes an industrial production index for the non-metallic minerals sector). That index could be a proxy for cement output. This is more about internal efficiency than demand, so not a top external driver.
- **Priority:** Low. (We mention for completeness but we will mostly rely on consumption and trade rather than separate production data, unless we want to model inventories.)

Cost Driver Data (Energy, Materials, CO₂, Logistics)

- **Source ID:** PT_OMIE_ELEC
- **Provider:** OMIE (Iberian Electricity Market Operator)
- **Dataset/Portal Name:** *MIBEL Day-Ahead Electricity Price – Portugal zone*
- **Link:** [OMIE – Market results](#) (and ENTSO-E Transparency platform for prices)
- **Description:** The **wholesale electricity price** in Portugal, which is crucial for a cement company's electricity cost (electricity is used in grinding and plant operations). Prices are set on the MIBEL

market (common with Spain), but there can be slight differences in PT zone. Extremely volatile in recent years due to fuel prices.

- **Key Variables: Daily spot price** (EUR/MWh) – we can use daily or monthly average. For example, the average price in 2022 spiked to ~**167 €/MWh** (record high) ⁴, whereas historically it was ~50 €/MWh. Currently (2025) it might be around 70 €/MWh ²⁷. We'll track the **monthly average price** and possibly peak prices.
- **Geography:** Portugal bidding zone (mostly equal to Spain's in day-ahead, but OMIE reports both).
- **Time Granularity & History:** **Hourly** prices (we aggregate to daily/monthly as needed). History since market start (~2007).
- **Update Frequency & Lag:** **Daily**, available immediately after market clearing (for next day's price). No lag practically.
- **Access Method:** OMIE website provides tables and a downloadable CSV of daily prices. ENTSO-E API also provides day-ahead prices by country. We can set up an automated fetch for daily data and compute monthly averages for our models.
- **Licensing/Cost:** Free (public market data).
- **Predictive Use Cases:** Directly used to **forecast electricity cost line** in financials. If our forecasting model includes cost of goods sold, we feed in projected power prices. Also indirectly, electricity price influences inflation and economic conditions (very high power prices can slow construction activity by squeezing contractor margins or causing project delays). But primary use is for **cost forecasting** – e.g., we might model that a +10€/MWh change in power price = +X € in monthly costs. For short-term forecasts, if we have futures prices, we can input those.
- **Leading/Lagging:** *Coincident for costs.* We know the prices as they occur and can forecast costs if we know our consumption. There's no lead for demand, except that extremely high sustained electricity prices could eventually dampen construction indirectly (through economy/inflation) – but that's more a second-order effect. We consider it leading for inflation and coincident for our expenses.
- **Predictive Relevance:** 5/5 (for cost forecasting) – Electricity is a significant cost for cement grinding and other processes, thus high relevance to EBITDA forecasting. In 2022, for example, failure to anticipate the price spike could cause major forecast errors in costs.
- **Ease of Integration:** 4/5 – Data is available and updated daily. The integration is straightforward via API or even a simple script. However, managing high-frequency data (hourly) might be overkill; we'll likely integrate monthly aggregates or weekly averages. That's easy to compute.
- **Comments/Caveats:** We might consider **hedging**: if the company has fixed-rate contracts or hedges, actual cost might not equal spot price moves in short term. But for forecasting, we often assume spot as a base. If needed, separate "price" and "consumption" forecasts for electricity (we may integrate energy consumption from internal data and multiply by external price). Also note the "Iberian electricity price cap mechanism" (in 2022–2023 Spain/PT had a gas cost subsidy affecting prices) – such regulatory changes can alter the relationship; but using actual price data captures all that.
- **Source ID:** NL_TTF_GAS
- **Provider:** ICE Endex / PEGAS (market operators) – aggregated by many sources (e.g., Bloomberg, EIA)
- **Dataset Name:** *Dutch TTF Natural Gas Price* (benchmark for EU gas)
- **Link:** [Trading Economics – Natural Gas EU](#) (for quick data) or [ICE reports] (subscription)
- **Description:** The leading natural gas price benchmark in Europe. Included because natural gas price heavily influences electricity price and is used directly if the company has gas-fired processes or uses

gas in captive power. In cement, direct gas use is limited (kilns use petcoke/coal/alt fuels), but gas drives power prices and some fleet fuel or drying processes.

- **Key Variables:** Daily/monthly gas price in €/MWh or USD/MMBtu. e.g., TTF price spiked to ~€180/MWh in Aug 2022, then dropped to ~€30/MWh by late 2023. We'd likely use a **monthly average TTF price** series.
- **Geography:** Europe-wide benchmark (Netherlands hub, but relevant for PT since gas markets are integrated).
- **Time & Lag:** **Daily**, real-time.
- **Access:** Free sources for historical (e.g., tradingeconomics API limited, or manual downloads; paid data from ICE for live). For forecasting, we might take futures curve values from public info.
- **Use:** Primarily to feed into power price forecasts or to model fuel cost if the company uses **natural gas for secondary heating or vehicles**. Also, it's a general economic indicator (gas price spikes can delay projects due to cost uncertainty).
- **Relevance:** 3/5 – moderate. We include it as part of energy cost modeling, but if our company doesn't consume gas directly, we might not need it separately beyond what's captured in electricity. Since it's an upstream driver for multiple costs, we monitor it.
- **Integration:** 3/5 – can get via web sources, but not as straightforward official API as electricity. We might decide to manually update this or use a data provider feed.

• **Source ID:** GLOBAL_COAL_PETCOKE

- **Provider:** World Bank Commodity Prices (for coal) / Industry price indices (Argus, Platts for petcoke – paid)
- **Dataset Name:** *Coal Price – Europe (API2), Petroleum Coke Price*
- **Link:** [World Bank Commodities “Pink Sheet”](#) (provides monthly coal price)
- **Description:** Coal and petcoke are the primary kiln fuels for cement in Portugal ²⁸ ²⁹ . Petcoke price data is not readily open, but it correlates with heavy fuel oil and coal prices. The World Bank publishes a free monthly **coal price** for Europe (API2 Rotterdam), and crude oil prices which can serve as proxies. These help forecast **fuel costs** – a major variable cost for clinker production.
- **Key Variables:** e.g., **Coal price** (USD/ton) – ~US\$150/ton in 2022, down to ~\$100 in 2023 (hypothetical). **Brent crude oil price** (USD/barrel) – influences shipping and petcoke (e.g., ~\$85 in 2025). **Petroleum coke price** – if available, often indexed to ~40-60% of Brent (depending on grade). If we get Argus or other data: e.g., Mid-sulfur petcoke USGC price in \$/ton.
- **Geography:** Global benchmark. (We assume import parity pricing for fuels.)
- **Time & History:** **Monthly** average prices (World Bank series goes back decades).
- **Update Frequency & Lag:** **Monthly**, with a short lag (WB releases commodity price data a few weeks after month-end). Daily prices are available from exchanges (ICE) for coal, but monthly suffices for forecasting horizon.
- **Access Method:** World Bank API (for coal, oil). Petcoke – if we don't have a paid feed, we might use a fixed relationship to oil or coal, or obtain quarterly updates from market reports.
- **Licensing/Cost:** World Bank data is free. Petcoke indices are typically proprietary (Argus, etc.), but one might find occasional public info or rely on supplier quotes.
- **Predictive Use Cases:** **Forecasting fuel cost line** (one of the “key cost drivers” explicitly mentioned). We will use coal/oil price to project **kiln fuel expenses**. For example, if we expect coal to increase 20%, we anticipate higher petcoke purchase prices and thus higher cost of clinker production. Also scenario analysis: high carbon scenario might drive up fossil fuel costs. In addition, fuel costs indirectly affect **logistics** (diesel) and **CO₂ costs** (higher coal use → more emissions permits needed).

- **Leading/Lagging:** *Coincident for costs.* Market prices immediately affect spot purchases, but companies often have contracts (lag effect if fixed quarterly). Still, as an external input, this is exogenous. Not directly related to predicting volumes (except extreme fuel costs could slow construction, but that's indirect).
- **Predictive Relevance:** 5/5 (for cost forecasting). Fuel is usually the **largest variable cost** in cement. Errors in fuel price assumptions can badly miss EBITDA. So it's tier 1 for cost.
- **Ease of Integration:** 4/5 – Getting *coal and oil* is easy via free sources. Petcoke specifically is 2/5 (harder), but we can manage with proxies or periodic updates. We might integrate a manual update process for petcoke (e.g., quarterly call with procurement or using Argus if we subscribe).
- **Comments/Caveats:** We should monitor **exchange rates** too (since fuel prices might be in USD but our accounts in EUR). Possibly include a EUR/USD series to convert costs. Also, quality differences: the coal price might not exactly match the calorific value cost of petcoke we use, but it's strongly correlated. We might model a fixed differential. If the company uses **alternative fuels** (biomass, waste), note those costs might behave differently (often lower cost or fixed gate fees). However, those usually reduce some petcoke consumption but not entirely decouple from market (e.g., biomass may follow different markets). For now, focusing on coal/petcoke covers the bulk of fuel cost risk.

• **Source ID:** EU_ETS_CO2

- **Provider:** European Energy Exchange (EEX) / ICE / EU Commission
- **Dataset/Portal Name:** *EU ETS Carbon Allowance Price (EUA)*
- **Link:** [Ember Climate EU ETS Price](#) (free viewer) or [ICE EUA Futures](#) (market data)
- **Description:** The price of CO₂ emission allowances under the EU Emissions Trading System – critically important for cement, which is a carbon-intensive sector. Cement producers must surrender EUAs for emissions, so the EUA price effectively acts as a “carbon tax” on each ton of cement (~0.6-0.8 tons CO₂ per ton cement ³⁰). The price has risen sharply in recent years ³¹, increasing costs.
- **Key Variables:** Daily **EUA price** (€/ton CO₂). As of 2025, prices are around €70–€90/ton ³². For instance, on 27 May 2025 the EUA price was **€71.8/ton** ⁵. We will likely use a monthly average EUA price for cost forecasting.
- **Geography:** EU-wide market (same price for all EU, including PT).
- **Time Granularity & History:** **Daily** market price. Historical series from 2005 (with phases; significant from ~2018 onward when price rose).
- **Update Frequency & Lag:** **Real-time** (daily settlements). We can incorporate data daily or monthly. No publication lag (market data).
- **Access Method:** EEX publishes settlement prices (but maybe behind login). Many financial websites publish daily EUA prices; the EU Commission's website often has monthly averages in reports. For automation, one could use an API from a data provider or scrape a site like Ember's (which might allow downloading historical prices). Since it's important, we might invest in an integration via, say, the ICE data (could be paid) or a third-party API.
- **Licensing/Cost:** Price data itself is not under strict copyright, but real-time access often requires subscription. However, end-of-day or delayed data is available in public sources. We can use those without cost.
- **Predictive Use Cases:** **Forecasting CO₂ compliance cost** – a direct component of our production cost. We will multiply our emission volumes by this price to estimate expense. It's vital for margin forecasting; at €70/ton, if our emission intensity is ~0.7 t CO₂/ton cement, that's nearly €50 cost per

ton – huge. Additionally, the trend of carbon price can influence long-term strategy (fuel switching, etc.), but in forecasting we mostly care for next 1-3 years scenarios of price. We could also use it in **scenario analysis** (e.g., what if carbon goes to €100?). Not typically used to predict sales volumes (except that extremely high carbon costs might raise cement prices and slightly dampen demand in the very long run).

- **Leading/Lagging:** *Coincident for cost.* We know the cost as price * emissions. No lead for demand. One could argue it's *leading for profit margins* – i.e., if we see futures indicating higher carbon prices, we know future costs will be higher, which could influence our pricing decisions or cost pass-through expectations.
- **Predictive Relevance:** 5/5 – **Very High** for financial performance forecasting. In a regulated industry, carbon cost can make or break the margin, especially as free allowances are phased out (noted to be gradually eliminated by 2030, meaning companies will pay the full carbon cost) ³³ ³⁴ . This data is non-negotiable to include.
- **Ease of Integration:** 3/5 – Moderate. We might need to get creative for an automated feed. Possibly use a source like Quandl or an open data project if available. Alternatively, manual monthly updates may suffice, since high-frequency precision isn't needed for board-level forecasts (but it is needed for internal tracking). We could set up a small script to scrape a known page daily.
- **Comments/Caveats:** Ensure to align price data with our compliance year – if we hedge or use forward contracts, the spot price might not reflect our actual cost. But generally, forecasting will assume spot or future prices. Also, note any government compensation or free allocation still received (if any) – though those are diminishing, they can offset gross cost. For forecasting, likely assume we eventually pay full price for each ton of CO₂ emitted, and thus this price multiplies by our emissions volume (which can be estimated from production volume and clinker factor). So, integration-wise, we need both the price and our emissions estimate (the latter from internal data or using clinker production forecasts and emission factors).

• **Source ID: PT_ERSE_FUEL**

- **Provider:** ENSE/ERSE (Energy Authority) or European Commission Oil Bulletin
- **Dataset Name:** *Retail Diesel Price – Portugal* (average)
- **Link:** [EU Oil Bulletin – Portugal](#) (weekly CSV for fuel prices by country)
- **Description:** Diesel fuel price in Portugal (average at the pump, incl. taxes). We care because of **logistics costs** – cement and concrete delivery by truck consumes diesel. Also, any on-site equipment uses diesel. A rising diesel price increases our distribution costs and can also impact customers (e.g., ready-mix companies' costs).
- **Key Variables: Diesel price (€/liter)** – e.g., currently ~€1.70/L for road diesel in PT. Historical: it was around €1.50 in 2021, spiked above €2 in 2022, etc. We might also consider **industrial diesel price (ex-tax)** if available, but pump price suffices as an indicator. The Oil Bulletin provides both with and without taxes. Also could track **fuel oil price** if relevant for some operations.
- **Geography:** National average (some variation regionally but minor).
- **Time Granularity & History:** **Weekly** data (which we can average to monthly). Decades-long history in EU Oil Bulletin (though tax regimes changed over time).
- **Update Frequency & Lag:** Weekly (usually Monday data published by Thursday). Very short lag.
- **Access Method:** European Commission provides a weekly CSV dataset. We can automate fetching the PT row. Alternatively, ERSE or DGEG might have a local source, but the EU one is straightforward.
- **Licensing/Cost:** Free and open.

- **Predictive Use Cases:** Use to **forecast transport and delivery costs** – a component of our distribution expenses and possibly cost-of-sales (if we deliver product) or SG&A (if we own a transport fleet). E.g., if diesel is projected to go up 10%, we anticipate higher freighting cost per ton (and maybe we adjust delivery fees accordingly). Also an **indicator of inflation**: diesel heavily impacts construction project costs (excavation, trucking of materials), which could influence project timelines (if fuel cost is high, contractors might slow work or demand contract adjustments). So high diesel costs can indirectly slow construction activity if prolonged. But primarily, we include it for **cost line forecasting**.
- **Leading/Lagging:** *Coincident*. Direct effect on costs when incurred. If we lock fuel contracts or use diesel hedging, that introduces a lag, but assuming we purchase regularly, the price changes flow through quickly. Diesel price is often correlated with oil price, which is more volatile daily, but weekly frequency is fine for our needs.
- **Predictive Relevance:** 4/5 – High for cost. Road transport is significant for cement (moving heavy product to customers) and aggregates/ready-mix operations. A large spike can materially impact margins if not passed on. However, in many cases, companies implement fuel surcharges or price escalators, so the net impact may be mitigated after a lag. Still, must be tracked.
- **Ease of Integration:** 5/5 – Very easy via the EU dataset.
- **Comments/Caveats:** We should decide whether to track **with taxes** (the pump price affects third-party transporters we hire) or **ex-tax** (if focusing on base cost). Probably pump price is fine because any haulier we use pays that price. Also note government interventions (fuel tax cuts or subsidies) can occur – but using the reported price captures those effects. If forecasting, one might link diesel price to oil price scenarios to project it forward.

• **Source ID:** PT_INE_CONSTR_COST

- **Provider:** Statistics Portugal (INE)
- **Dataset Name:** *Índice de Custos de Construção de Habitação Nova (ICCHN)* – Construction Cost Index for New Housing
- **Link:** [INE – Construction Cost Index Press Release](#) (example for June 2025) ³⁵ ; data also on INE database.
- **Description:** Index tracking the evolution of construction costs for new residential buildings, broken into **Materials** and **Labor** cost sub-indices. This is an official measure often used for contract escalation. It reflects broad input cost trends in construction (material prices like cement, steel, etc., and construction wages). For a cement company, it provides context on cost inflation that could affect our **input costs (materials like aggregates, packaging, etc., and subcontractor costs)** and also our customers' costs (which might affect demand or pricing environment).
- **Key Variables:** Overall Construction Cost Index (2015=100 or base 2021=100). Also **Materials Cost Index** and **Labor Cost Index** components. For example, *July 2025: construction costs +4.8% YoY (labor +8.9%, materials +1.5% YoY)* ³⁶ . This indicated labor cost acceleration while materials cost inflation was modest. We also get monthly % changes and moving averages.
- **Geography:** Mainland Portugal (Continente) – since it's focused on mainland housing (Azores/Madeira might be separate but usually the index is just for continent).
- **Time Granularity & History:** **Monthly** index. History: available at least since 2000 (with base changes; older base 2000 index available ³⁷ , now updated to base 2021).
- **Update Frequency & Lag:** **Monthly**, released ~6 weeks after month (e.g., June data in early August ³⁵).

- **Access Method:** INE API (Indicator code 0011751 for ICCHN% changes ³⁸). Also INE's highlights and Excel files. The data is easily accessible.
- **Licensing/Cost:** Free.
- **Predictive Use Cases:** Helps in **forecasting our own cost inflation** for inputs not covered by specific commodity indices. For example, it captures the general rise in construction materials (which includes not just cement but also steel, lumber, etc.) – relevant if we purchase such items or for maintenance CAPEX costs. It also is a gauge for **wage pressure** in the construction sector, which can translate to our labor costs (e.g., if construction wages are up 9%, our plant maintenance contractors or drivers may demand more pay). Additionally, because cement prices often move somewhat with overall construction costs (e.g., contractors might tolerate higher cement prices if overall costs are rising anyway), it can inform pricing strategy. We might use it to escalate internal budgets or to justify price adjustments (some contracts might even link to this index). For forecasting models, it's an exogenous input to project **cost line items like maintenance, outsourced services, minor raw materials**.
- **Leading/Lagging:** *Coincident to slight lag*. It reflects past month's costs. It's somewhat **lagging** as an economic indicator (it goes up after underlying commodity/wage increases have occurred). But it's useful as a summary of multiple cost drivers. There is no "leading" property in it per se, but a steady rise here can validate that cost pressures are sustained, meaning we may forecast margin pressure if we can't raise prices equivalently.
- **Predictive Relevance:** 3/5 – **Medium**. It's a composite, so not as pinpoint as specific energy or material prices. But it's very relevant qualitatively (e.g., in 2021-2022 this index jumped due to material shortages, which signaled margin squeezes industry-wide). We will monitor it in Tier 2. For specific forecasting, we lean on individual drivers (energy, etc.), but this index ensures we don't miss the forest for the trees.
- **Ease of Integration:** 5/5 – Easy via INE.
- **Comments/Caveats:** The weight of materials vs labor is roughly 50/50 in this index. For a cement producer, labor cost in construction industry might not directly equal our labor cost changes (different sectors, but correlated due to economy-wide wage trends). The materials part includes many things (steel, cement itself, etc.). We should not use this index to double-count cement's own price – rather, we exclude cement when analyzing customers' cost structure since we are the cement supplier. However, as a high-level indicator, it's fine. Also note this index is for *housing* – for infrastructure projects, cost dynamics might differ (e.g., more steel, more asphalt etc.). But housing is the bulk of private construction.
- **Source ID:** PT_IPA_WEATHER
- **Provider:** IPMA – Instituto Português do Mar e da Atmosfera (Portuguese weather service)
- **Dataset Name:** *Meteorological Series (precipitation, temperature) – Portugal*
- **Link:** [IPMA Long Series Download](#) (open data for monthly climate indicators) ³⁹ ⁴⁰
- **Description:** Historical weather data such as **monthly rainfall, mean temperature, and extreme weather days**. Weather significantly affects construction: heavy rain can halt outdoor concrete pours and earthworks; extreme heat can also slow work. Thus, deviations from normal weather cause short-term variance in cement demand (seasonality). For example, an unusually dry winter can boost construction activity, whereas a very wet month will depress it (as noted by INE for March 2018 where rainfall conditioned construction activity ⁷). We incorporate weather data to adjust and improve short-term forecasts.

- **Key Variables:**
 - **Total precipitation (mm)** by month, by region (or national average).
 - **Number of rainy days** per month.
 - **Average temperature** (which can matter in extreme heat or cold).
 - Possibly **deviation from normal (1971-2000 average)** provided by IPMA for each month ⁴¹. For instance, March 2018 had record rainfall ~2-3 times the normal, correlating with a construction output drop ⁷.
 - We can derive indicators like “months with rainfall > X mm” or use cumulative rain as a negative predictor for cement sales in that month.
- **Geography:** Available by region/city (IPMA has station data, and aggregated “Portugal Mainland” series). For forecasting, a national aggregate or separating North vs South (since weather patterns differ) could be useful if our sales are regional.
- **Time Granularity & History:** **Daily** station data, aggregated to **monthly**. Long history (for Lisbon, etc., since 19th century; but for our purpose, last 10-20 years monthly is enough).
- **Update Frequency & Lag:** **Monthly** climate summaries are published a few days after month-end (IPMA issues a monthly climate bulletin). The open data sets might update with some lag for quality control, but typically within a month. We don’t need real-time; even a slightly lagged actual can be used to adjust the model retrospectively (or near-real-time weather can be pulled from global API for the ongoing month).
- **Access Method:** IPMA provides an open API for forecasts and some data ⁴², and a download for historical series ³⁹. Likely, we can download a CSV of monthly precipitation and temperatures by region. Automation might require hitting their site or using a provided API endpoint. Alternatively, global sources like NOAA or ECMWF reanalysis could give similar data. But IPMA is authoritative locally.
- **Licensing/Cost:** Free for use with attribution (open government data policy) ⁴³.
- **Predictive Use Cases:**
 - Adjusting near-term forecasts:** If a particular quarter is significantly wetter or drier than average, we adjust volume projections. For example, if our model baseline assumes normal weather, but April saw 50% more rain than usual, we might reduce the April sales forecast accordingly (and possibly expect a rebound later as pent-up work is completed). Also useful for
 - explaining variance:** to management, we can quantify how much of a miss in sales was due to weather vs other factors. On a longer horizon, incorporating seasonal averages helps ensure our forecast has the right seasonal pattern (peak in dry summer, slow in rainy winter). We could also use predictions (seasonal forecasts if available) to anticipate a harsh winter or hot summer and tweak scenarios.
- **Leading/Lagging:** Weather itself is *contemporaneous*, but in forecasting we mostly use it as a known input for the recent past or current period. It’s not leading beyond a very short term (weather forecasts ~1-2 weeks out are reliable, seasonal forecasts are probabilistic). We will not rely on long-term weather predictions due to uncertainty, except perhaps to scenario-test (e.g., an El Niño year might mean wetter winter).
- **Predictive Relevance:** 4/5 – **High (short-term)**. Weather is often the **main cause of short-term fluctuations** in cement demand that cannot be explained by economic factors. For a 1-3 month horizon, getting weather adjustments right can greatly improve forecast accuracy. Over 12+ months, weather averages out (so for multi-year trend it’s less critical). But given the question’s scope includes monthly/weekly forecasting, this is very relevant. In Portugal’s climate, rainfall in winter is the key constraint (unlike places with freezing temperatures).
- **Ease of Integration:** 4/5 – Data itself is straightforward. Need a bit of setup to fetch and aggregate. Possibly some cleaning of station data if we go that route. But IPMA’s “series longas” provides already aggregated series (like national precipitation anomaly) which simplifies integration.

- **Comments/Caveats:** We should correlate historical sales with rainfall to calibrate the impact (e.g., X mm above normal = Y% drop in sales). This relationship may be non-linear (e.g., light rain has little effect, but heavy rain has big effect). We might create a threshold variable (rainy days > some mm). We should also account for timing (rain in early month vs late). But for our purposes, monthly totals and anomalies are a good proxy. Also consider extreme heat in summer – if there are heatwaves, labor productivity drops. IPMA provides info on heatwaves as well. If climate change trends continue, we may want to incorporate gradually shifting seasonality (e.g., longer dry seasons, etc.), but that's beyond short-term forecasting – more for scenario planning.

- **Source ID:** EU_ECON_SENTIMENT

- **Provider:** European Commission – DG ECFIN
- **Dataset Name:** *Economic Sentiment Indicator (ESI)* and **PMI** (purchasing managers' index) – *Composite indicators*
- **Link:** [DG ECFIN ESI](#) (for ESI); [S&P Global PMI Press Release] (for Portugal PMI if available)
- **Description:** Broad indicators of economic climate. The ESI combines confidence across sectors (industry, services, consumers, construction, retail) into one index. PMI, if available, is a monthly index (above 50 = growth) usually for manufacturing or services; sometimes a construction PMI exists for large economies but not sure for PT specifically. These indicators capture general economic momentum and can be leading for the construction sector indirectly (if economy is doing well, public finances improve, private investment confidence rises, etc.).
- **Use:** These are **optional** but can provide an extra leading signal or validate what more specific indicators say. For instance, if all construction indicators look good but overall sentiment is plunging (maybe due to external crisis), we might moderate our forecast anticipating a broad slowdown. Conversely, strong sentiment might support higher end of forecast range. They can be input features in an ML model as broad regressors.
- **Priority:** Low/Medium (Tier 2). Construction-specific data is more direct. We'd use ESI/PMI as supplemental.
- **Integration:** Easy (ESI from EU API, PMI from news).
- **Caveats:** ESI includes construction confidence which we already separately have; PMI may mostly reflect manufacturing, which indirectly affects cement (through commercial/industrial building demand). So not primary.

Recommended Tier 1 & Tier 2 Data Stack

Tier 1 – Core Sources to Integrate First: These have **high predictive value** for our financial KPIs and are reasonably accessible. We should prioritize building automated pipelines for these:

- **Building Permits (INE)** – *Predictive relevance:* Very high for future sales volumes ¹. *Ease:* API available. **Action:** Pull monthly permit counts (dwellings, area) via INE API. Update frequency: monthly/quarterly; integrate with ~1 month lag. Use to forecast volume by region.
- **Cement Consumption (ATIC/AICCOPN)** – *Relevance:* Ground truth of market demand ¹⁶. *Ease:* Moderate (need to scrape or source via industry). **Action:** Obtain monthly consumption from AICCOPN bulletins or ATIC directly. Ingest by end of following month. Use to track market and validate volume forecasts.

- **Construction Output Index (INE)** – *Relevance:* High (current activity indicator) ⁹ . *Ease:* Easy (API). **Action:** Pull monthly; refresh ~45 days after month-end. Use for nowcasting current quarter volumes (as a coincident check).
- **Mortgage New Loans & Interest (Banco de Portugal)** – *Relevance:* High leading indicator for housing demand ³ . *Ease:* API. **Action:** Integrate monthly via BPstat API (with ~30-day lag). Use new housing credit growth as a predictor for permit and sales upticks; use interest rates to gauge demand outlook (rising rates = possibly slower future permits).
- **Electricity Price (OMIE/MIBEL)** – *Relevance:* Very high for energy cost forecasting ⁴ . *Ease:* API/CSV available. **Action:** Set up daily fetch; compute monthly average price. Refresh daily or monthly. Feed into cost model to update power cost projection monthly (or even weekly in volatile times).
- **Coal/Petcoke Price (proxy via World Bank coal)** – *Relevance:* Very high for fuel costs. *Ease:* API. **Action:** Pull monthly coal price USD/ton, convert to EUR and adjust if needed for petcoke. Update monthly. Use for kiln fuel cost forecast; adjust if company provides specific petcoke price inputs.
- **CO₂ EUA Price** – *Relevance:* Extremely high for cost ³² . *Ease:* moderate, but critical. **Action:** Use a reliable source (Ember or manual update from market data) to get daily/monthly price. Update at least monthly (more if big swings). Multiply by emission volume to project carbon costs. Possibly integrate futures for scenario (e.g., assume €80/ton next year).
- **Diesel Price (EU Oil Bulletin)** – *Relevance:* High for transport cost. *Ease:* CSV from EU. **Action:** Download weekly, take monthly avg. Automate via Python script. Use to update distribution cost assumptions monthly.
- **Weather (IPMA rainfall)** – *Relevance:* High for short-term volume forecast ⁷ . *Ease:* moderate (but one-time setup). **Action:** Integrate monthly precipitation anomaly (national or by region). Update shortly after month-end. Use in regression model to adjust monthly sales forecasts (e.g., incorporate as independent variable). Also use current month's weather forecast to adjust near-future (if heavy rain predicted, temper expectation).
- **Public Works Contracts (Portal Base/AICCOPN Barometer)** – *Relevance:* High for infrastructure demand pipeline ⁶ . *Ease:* Moderate (API with registration or scraping AICCOPN summary). **Action:** In near-term, use AICCOPN Barometer monthly summary (published ~T+25 days) to get values: total tenders and contracts YTD and YoY%. Eventually, consider using Portal Base's API to get monthly aggregated values of public works tenders and awards (with IMPIC authorization). This would refresh monthly. Use this to forecast cement demand from infrastructure: e.g., if public contract awards jump, expect higher volumes in coming quarters. For now, since integration is tricky, we might manually pull this data for modeling until automated solution is in place.
- **Construction Cost Index (INE)** – *Relevance:* Medium for cost inflation monitoring ³⁶ . *Ease:* Easy (API). **Action:** Pull monthly (6-week lag). Use material vs labor sub-index to update assumptions on maintenance/materials costs and wage inflation in our cost base. Tier1 for awareness (since we need to manage cost expectations), though for strict forecasting it's secondary to direct commodity prices.

Using these Tier 1 sources, we cover the most critical drivers: permits/credit (demand), consumption (actual market size), energy/fuel/CO₂ (costs), and major market signals (public works, weather).

Tier 2 – Additional Sources for Later Integration: These are useful but not as urgent or with lower ROI initially:

- **House Price Index (INE)** – Important for context but quarterly and lagging; can be added later for scenario modeling (Tier2).
- **Bank Appraisal Values (INE)** – Already somewhat covered by permits & credit, but could add for finer predictive power in housing cycles (Tier2).

- **Construction Confidence (INE/EC)** – We might use it qualitatively early on; integrating numeric series can come later once core model is built (Tier2). It can improve turning-point detection.
- **Regional Real Estate Indicators (Conf. Imobiliário)** – Only if we need more granular forecasts (Tier2, paid).
- **Trade Data (INE/Eurostat) for cement** – We should incorporate this when we want to model **market share or detect import shocks**. It's slightly lower priority as current imports are relatively low, but if our strategy shifts or imports rise, this becomes important. Could schedule to fetch trade data monthly (Tier2).
- **Cembureau/EU aggregates** – For benchmarking and long-term planning, not needed in automated pipeline (manual annual check suffices).
- **Other Macro (ESI, PMI, GDP)** – These can be integrated into advanced forecasting models (e.g., a VAR model including GDP or ESI). Initially, we rely on more direct indicators. We can add macro variables later to capture any residual variability (Tier2).
- **Alternate fuel usage or costs** – If we start using significant alternative fuels, we might track indices for biomass or waste fuel costs (not urgent now; Tier2 if needed).
- **Labor market** – e.g., construction employment or wages beyond what's in cost index. Could integrate if labor becomes a bottleneck issue (Tier2 watchlist).
- **Climate forecasts** – e.g., seasonal forecast from IPMA if available (to anticipate a severe winter). This is more experimental; we might consider it later.

Justification: Tier 1 sources are those that either have a *direct causal relationship* with our financial outcomes (e.g., permits → sales, fuel price → costs) or represent the **actual performance metric** we predict (cement consumption). They also tend to have relatively low integration effort for high reward (APIs, well-structured data). Tier 2 sources, while useful, either overlap with others or require more effort for incremental benefit. We will implement Tier 1 first to get the core predictive system running; Tier 2 can be layered on to refine and edge-case the model.

Tier 1 Integration Plan (high-level):

- **API-based feeds (easy automation):** INE (permits, output index, cost index, confidence), Banco de Portugal (credit, rates), Eurostat/EU (diesel price, possibly trade), OMIE/ENTSO-E (electricity). For these, set up scheduled API calls or data downloads, e.g., a monthly job for INE/BdP data at month + 1 (when data is released), a weekly job for energy prices (or daily for power). Data will be ingested into our data lake or time-series database with timestamps and unique IDs.
- **Scraping/manual (requires process):** ATIC cement consumption – assign a team member or script to retrieve AICCOPN's PDF each month around the 25th and parse the relevant figures (could use PDF text scraping or possibly request data from ATIC). Similarly, AICCOPN's Public Works Barometer – scrape the summary text for percentages and values ⁶. In parallel, work on obtaining API access to Portal Base for a more automated solution to public tenders (this might be a project with IT, given the need for IMPIC approval).
- **External market prices:** CO₂ and coal/petcoke – subscribe to a data source if needed (there are free daily updates on some financial sites for CO₂; coal via World Bank monthly is fine). We can schedule a script to fetch the latest CO₂ price weekly from a known source (like ICE via an unofficial API or a site like Ember's JSON if provided). Alternatively, input manually at month-end from a market report – since our forecasting is not trading, a small manual step is acceptable initially. Over time, formalize via a data vendor API if accuracy is paramount.
- **Weather:** Use IPMA's open data – likely download the monthly climate summary data file when available (they have a static link or an API as per documentation ³⁹). Could also use an automated

retrieval from NOAA as backup (e.g., NOAA CPC has global precip indices). Integrate this data alongside sales data to generate weather adjustment factors.

Refresh Frequencies:

- **Daily/Real-time:** Electricity price (daily ingest, as it's volatile), CO₂ price (could be daily or at least weekly due to volatility). These feed into rolling cost forecasts.
- **Weekly:** Diesel price (from weekly bulletin). Possibly coal/oil markets if we want more frequent updates than monthly.
- **Monthly:** Building permits, cement consumption, credit data, production index, construction cost index, weather (all monthly or monthly-updated). We'll schedule these updates aligned with their release calendar: e.g., INE releases often occur mid-month for the prior month – we'll mark those dates. Banco de Portugal around end-of-month for previous month's loans. ATIC data by ~25th. Public works (AICCOPN) around 25th. We can have an "update day" each month where the pipeline fetches all newly available monthly indicators and updates the forecasting model inputs.
- **Quarterly:** House price index (we can incorporate when available each quarter, but it won't be a core part of short-term forecast, more for medium-term scenario update).
- **Annual:** Any high-level benchmarks (Cembureau) just for reference, not automated.

By structuring ingestion this way, our data pipeline will continuously collect and update the key drivers. The forecasting model (which is outside this scope) can then pull the latest values and project financials (volumes, revenue, costs) with these externals driving the projections.

Integration & Next Steps

Phasing the Integration: We will start by building the pipeline for Tier 1 sources. A sensible phase 1 is to get **INE and BdP API data flowing**, since those are straightforward. In parallel, set up **energy price scrapers/APIs** (electricity, diesel, etc.). The more manual sources (cement consumption, public works) we can initially handle with a semi-manual process (e.g., an analyst enters the number each month from the report) to not delay model development – but aim to automate them in phase 2 either through web scraping or data-sharing agreements. Tier 2 sources can be added once the core model is running, to incrementally improve it.

Data Quality & Alignment Issues:

- **Frequency mismatches:** We have monthly, quarterly, weekly, daily data. We'll need to align these in the model. Generally, we will convert everything to monthly frequency for forecasting monthly financials. (Daily power price becomes monthly avg, weekly diesel to monthly avg, etc.) For weekly or daily data, ensure proper averaging or end-of-period selection so as not to bias the month (e.g., since weeks don't align perfectly with months, use a weighted average if needed).
- **Timing and lags:** Many indicators are published with a lag (permits quarterly, etc.). Our forecast may need the latest partial info – e.g., by mid-quarter we might only have permits for the previous quarter. We might use proxies or high-frequency clues (like confidence or internal order intake) to fill the gap. We should also be mindful of revision schedules (e.g., INE might revise a previous month slightly). We will incorporate a process to update prior data if revisions occur (the APIs typically give the latest revised values).

- **Geographical alignment:** Our financial forecasting might be segmented (by region or product). External data often is national totals. We should be careful matching them. For example, if we forecast regionally, we'd need regional permits data. If unavailable, using national as proxy might reduce precision for regional splits. A next step could be to obtain regional breakdowns (INE permits by region, etc.). Initially, we focus on national-level forecast of total volumes and costs, then distribute internally.
- **Data coherence:** Some indicators overlap (e.g., permits and confidence both gauge future activity). We must avoid multicollinearity in models – perhaps use one primary and keep others for cross-validation. Also ensure consistency: e.g., if permit growth says +10% and credit +40%, it implies a robust scenario – if they diverge, investigate why (maybe one-off factors).
- **Seasonal adjustment:** Some data are seasonally adjusted (construction index ⁴⁴), others are raw (cement consumption). For modeling, we might want to use seasonally adjusted versions or explicitly model seasonality. We should check if INE provides seasonally adjusted permits (likely not, since they are irregular). We might perform our own seasonal adjustment on certain series to identify underlying trends. Alternatively, incorporate seasonal dummies in the regression.
- **Unit and scale differences:** We'll standardize units (e.g., fuel in €/ton vs €/MWh conversions, currency conversion for global prices, etc.). Also, make sure to express variables in a model-friendly way (often YoY % changes or indexed values for stationary modeling). For instance, we might use permit growth rate rather than absolute permits in regression to avoid non-stationarity.

Testing Incremental Value: Once integrated, we will conduct back-tests: e.g., build a simple model predicting cement consumption using permits, credit, confidence, etc., and see which combination yields the best out-of-sample predictions. We expect permits and credit to significantly improve forecast accuracy (leading info), and weather to reduce error in volatile months. We'll also test cost forecast accuracy by feeding actual price data and comparing to actual cost outcomes, ensuring our assumptions (e.g., fuel per ton, etc.) are calibrated.

Next Steps Summary:

1. **Set up data pipeline** for Tier 1 sources – leverage APIs where possible (INE, BdP, EU), and scripts for scrapes (AICCOPN, OMIE if no API). Establish a data storage (SQL or time-series DB) with tables for each indicator keyed by date.
2. **Populate historical data** – gather at least 5-10 years of history for each series (for modeling purposes). E.g., get permits back to 2000s, cement consumption history (maybe from ATIC or historical stats) – might need manual entry for older years from published reports. The richer the history, the better for model training and understanding cyclicalities.
3. **Develop forecasting models** (separate effort) that use these external inputs – likely a combination of regression models for volume (e.g., multiple regression or machine learning with permits, credit, etc. as features) and cost models (bottom-up multiplication of inputs and prices). Use cross-validation to see which indicators truly add value versus noise.
4. **Dashboard & Monitoring:** Create a dashboard that shows the latest values of all integrated indicators vs their historical range, with alerts on extreme moves. This helps the team qualitatively assess situations (e.g., “construction confidence just fell to a 5-year low – check if our forecast needs a downside adjustment”).
5. **Tier 2 integration:** After validating Tier 1 model accuracy, integrate Tier 2 data one by one to see if they significantly reduce forecast error or provide additional insight. For instance, add house price index to see if it improves multi-year volume trend forecasting; or import volumes to adjust market

share assumptions. If an indicator doesn't materially improve the model, we can keep it as a monitor rather than a core input.

6. **Stay updated:** Many sources have annual base updates (INE might rebase indices every 5-10 years, EU might change methodology, etc.). We need to maintain the pipeline accordingly (e.g., switch API indicator codes when INE updates the base year, as happened with the construction cost index ⁴⁵). Also, maintain a calendar of release dates for planning forecasting cycles (for instance, knowing that permit data for Q4 is only out in early Feb means our January forecast for Q1 has to use partial info or proxies).
7. **Data governance:** Document each data source (metadata like last updated, source link, transformation applied) so that the AI team and finance team can trust and understand the inputs. Possibly implement version control on data (so if sources revise history, we know what changed).

By following this plan, the cement company will have a robust external data infrastructure underpinning its financial forecasting. We will be able to anticipate market shifts (using permits, credit, sentiment) and react quickly to cost fluctuations (energy, carbon) with data-driven insights, improving both the accuracy of forecasts and the agility of decision-making.

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